



WARWICK

Advanced Computer Graphics

Module Introduction

Dr. Tom Bashford-Rogers

WM9M3

▶ Advanced Computer Graphics

▶ Welcome!



Introduction

- ▶ The aim of this module is for students to be able to develop advanced theoretical concepts in terms of mathematics and algorithms in computer graphics that will serve them in good stead for future developments in the area.



Who

- ▶ Module Lead: Dr Tom Bashford-Rogers
- ▶ Office Hours: By appointment
 - My office: Room 360 in IMC



Objectives

- ▶ Have a comprehensive knowledge of physically-based rendering and shading
- ▶ Reduce complex formulations of physically-based solutions for high performance in interactive environments
- ▶ Review and critically evaluate state-of-the-art computer graphics research



Topics

- ▶ *Ray Tracing*
 - ▶ *Lighting*
 - ▶ *Monte Carlo Integration*
 - ▶ *Light Transport and Materials*
 - ▶ *Physically Based Rendering Algorithms*
 - ▶ *GPU Raytracing and Approximations*
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Topics

- ▶ As before, this is a big subject
 - We will go quickly
 - There will be large amounts of information
- ▶ Will be code in most topics
 - MUST do the code when asked
 - Limited time in lectures, complete outside lectures

Code

- ▶ The module is based on a skeleton of code
 - I have taken into account feedback from CG
 - You have structure in place
 - The code is pedagogical, not aimed at performance
 - Feel free to re-write it!



Assessment

- ▶ One component
 - Code (85%)
 - Report (15%)
 - 60 hours work
- ▶ Include a video like usual



Assessment

▶ Code

- Builds on what you develop in the lectures
- You **must** complete code when asked
 - Use time between lectures if you don't finish in the lecture



Assessment

▶ Code

- Tackle easier assessment tasks first
- Marking is not all or nothing
 - If you tried something and it didn't work, (briefly) mention it in the report and point to commented out code in the repo. You might still get marks.
- Include a github repo link in your report



Assessment

► Report

– Use LaTeX on Overleaf

- Use the LaTeX template on Moodle

– Report should be short and to the point

- 500 – 1000 words max
- Follow the structure from the assignment specification, include maths, graphs and images



Assessment

- ▶ Code has two parts:
 - Part 1: Extend the path tracer we write in the lectures
 - Part 2: Implement one or more other rendering algorithms
- ▶ Part 1 is easier





Questions?

Computer Graphics

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